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Renovation of Optically Active Phenanthrolines as Powerful Chiral Ligands for Versatile Asymmetric Metal Catalysis

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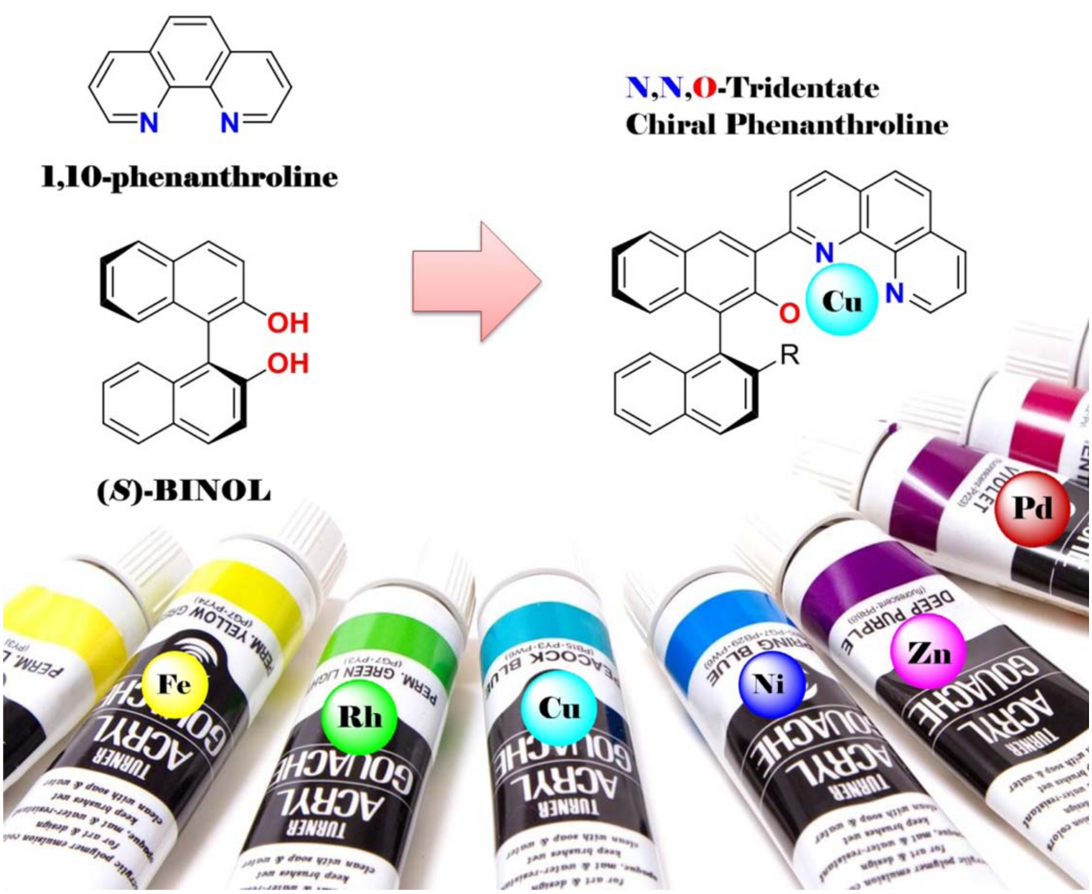

ABSTRACT: In the field of asymmetric synthesis, the development of new chiral ligands has been regarded as an attractive challenge for decades. Novel chiral ligands can often have a great impact on synthetic protocols. In this context, we are currently interested in the application of 1,10-phenanthroline (phen) as an entirely new class of chiral ligand. To handle this issue, we designed a chiral phen ligand that provides the *N*,*N*,*O*-tridentate coordination of the phen moiety and an additional phenolic hydroxyl group. As phen possesses greater coordination ability with various ions, our chiral phen ligand would be valuable as one of the "privileged" chiral ligands applied to a broad range of metal catalysts and new reactions. This account summarizes the results of the application of the chiral phen ligand to various kinds of metal catalysis.

Keywords: asymmetric synthesis, chiral ligands, homogeneous catalysis, transition metals, tridentate ligands

1. Introduction

The development of new classes of chiral ligands has been regarded as one of the most fundamental and important research topics in asymmetric synthesis. Our research group is currently interested in the use of 1,10-phenanthroline (phen) as a backbone for chiral ligands.[1] The phen backbone is one of the most classical and widely applied chelate reagents and has two *cis*-oriented nitrogen atoms (Figure 1).[2] Its strong affinity with a wide range of metal ions, including not only transition metals but also typical elements, is due to the rigid and planar structure containing the two *cis*-oriented nitrogen atoms, thus enabling the formation of entropically favored strong complexes compared to structurally analogous 2,2-bipyridine ligands.[2]

Therefore, phen has also been employed as an effective ligand in various reactions catalyzed by transition-metal complexes.[3] However, it is a somewhat surprising fact that chiral ligands utilizing the phen backbone have been rarely studied, even though there is the potential for broad applications and versatile performances with transition-metal catalysts.[1,4] The first chiral phen, with a simple (*R*)-*sec*-butyl group at the C-3 position, was designed by Gladiali et al. in 1987. The catalytic activity was evaluated in Rh-catalyzed enantioselective transfer hydrogenation of ketones, albeit with low enantioselectivity (Scheme 1).[5a]

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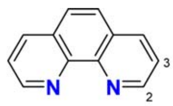
This work is dedicated to the 15th anniversary of the Nobel Prize in Chemistry for Professor Ryoji Noyori, awarded in 2001 together with W. S. Knowles and K. B. Sharpless

Although this survey surely confirmed the applicability of chiral phen in asymmetric synthesis, the following project on the development of chiral phen ligands has been less explored.[5–9] The major studies with these chiral phen ligands have been limited to (i) Rh-catalyzed reduction,[5] (ii) Cu-catalyzed cyclopropanation,[6a,b] (iii) Cu-catalyzed allylic oxidation,[6c] (iv) Cu-catalyzed Henry reaction and amination reaction,[6d,e] (v) Pd-catalyzed Tsuji–Trost reaction,[7] (vi) Sc or In-catalyzed ring opening of *meso*-epoxides,[8] and (vii) Fe-catalyzed epoxidation.[9] We assumed that this is due to the inherent difficulty of structural modification of phen, which lacks sp^3 carbons at which a chiral center can be introduced. Therefore, the basic strategy to design chiral ligands commonly relies on the use of central or axial chirality to establish the effective chiral environment. However, previous examples of chiral phen ligands were mainly based on the attachment of pendent chiral auxiliaries at the C-2 or C-3 positions of the phen backbone (Figure 2).[5–8] In addition, most of these chiral side chains were derived from natural chiral sources and the difficulty in structural manipulations leads to a lack of molecular accuracy. One exception of natural chiral auxiliary-free phen ligands was reported by Nishikawa and Yamamoto.[9]

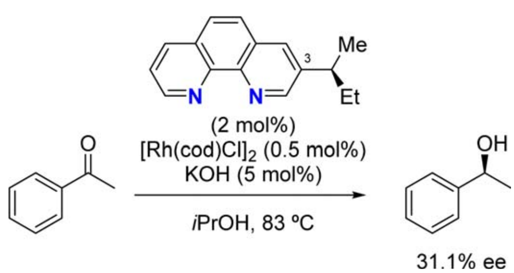
Another difficulty in the design of chiral phen ligands with pendent chiral auxiliaries is controlling the conformational fixation. The C–C bond between phen and the chiral auxiliary rotates freely and the attainment of the appropriate and effective chiral environment is basically troublesome (Figure 3a). We envisioned that design of tridentate chiral phen ligands bearing the other coordinative chiral auxiliary would enable the formation of more stable complexes (Figure 3b).

In this context, we recently designed the novel chiral phen ligand (*S*)-**1** with axial chirality provided by the binaphthyl backbone (Figure 4).[11] The significant difference from previous chiral phen ligands is the introduction of a phenolic hydroxy group, which provides not only classical *N*,*N*-



1,10-Phenanthroline (phen)
* *N,N*-Bidentate neutral ligand
* *cis*-Oriented two nitrogen atoms
* High coordination ability with various metal ions
* Rigid and planar structure based on sp^2 carbon
* No inherent chirality

Fig. 1. 1,10-Phenanthroline (phen).



Scheme 1. Pioneering report of a chiral phen ligand by Gladiali.

bidentate but also *N*,*N*,*O*-tridentate coordination with a variety of metal ions. The tridentate complex will be more entropically stabilized in combination with the presence of the C–O covalent bond, and the catalytic activity might be distinguished from that of the corresponding bidentate complexes. Synthetic manipulation of binaphthyl compounds such as BINOL has been well established[12] and the fine tuning of substituents in ligand (*S*)-**1** realizes the highly designable ligand structure.

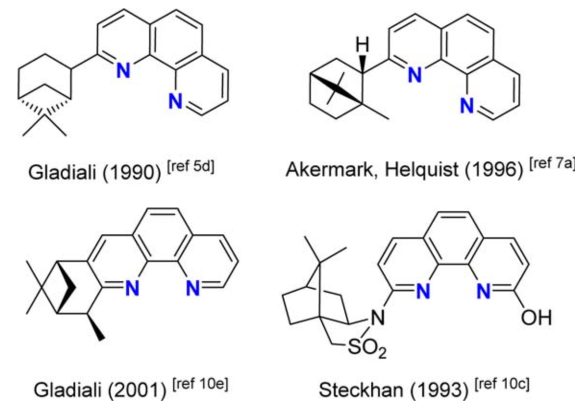


Fig. 2. Previous examples of chiral phen ligands with pendent natural chiral sources.

2. Synthesis of Axially Chiral *N*,*N*,*O*-Tridentate Phen Ligands

The synthesis of chiral phen (*S*)-**1** starting from commercially available (*S*)-BINOL is shown in Scheme 2. Preparation of the reported compound (*S*)-**4** was carried out by the modified procedure reported by Katsuki.[13] The initial step is the preparation of monotriflate (*S*)-**2** with Tf_2O and *N*,*N*-diisopropylethylamine. Nickel-catalyzed Kumada cross-coupling of (*S*)-**2** with Grignard reagents provided compounds (*S*)-**3** having a variety of aromatic substituents. The hydrogenation of (*S*)-**2** with Pd/C gave (*S*)-**3** without any substituent at the R position (R = H).[14] The protection of the hydroxyl moiety with the MOM protecting group furnished (*S*)-**4**.

The reaction of *ortho*-lithiated (*S*)-**4** with phen gave the corresponding adduct (*S*)-**5**. The oxidation of (*S*)-**5** with excess

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Hisao Nishiyama was born in 1951 in Japan. He received his Bachelor and Master of Engineering under the direction of Prof. Yoshio Ishii from Nagoya University in 1973 and 1975, respectively. He received his Dr. Sci. from the Tokyo Institute of Technology in 1980. He worked at the Research Centers of Toray Industries Inc. from 1975 to 1980, and then he worked at Toyohashi University of Technology from 1980 to 2002, where he was associate professor from 1985 and full professor from 1996. Since September 2002, he has been a professor of Nagoya University. His main research interests are homogeneous catalysis, asymmetric catalysis with nitrogen-based ligands and transition metals, and organometallic chemistry. He received the Progressive Award of Young Chemists from the Chemical Society of Japan in 1984 and the Award of Organic Synthetic Chemistry of Japan in 1996 and 2015.

amounts of manganese(IV) oxide and the final deprotection of the MOM group of (*S*)-**6** under acidic conditions provided the desired axially chiral *N*,*N*,*O*-tridentate phen ligands (*S*)-**1**. The molecular structure of (*S*)-**1c** bearing a *p*-tolyl group was confirmed by X-ray crystallographic analysis (Figure 5).[11]

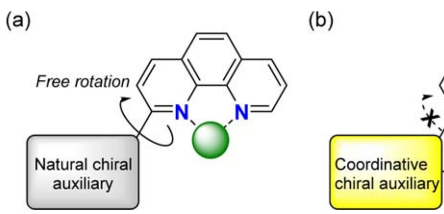


Fig. 3. Design concept for chiral phen ligands.

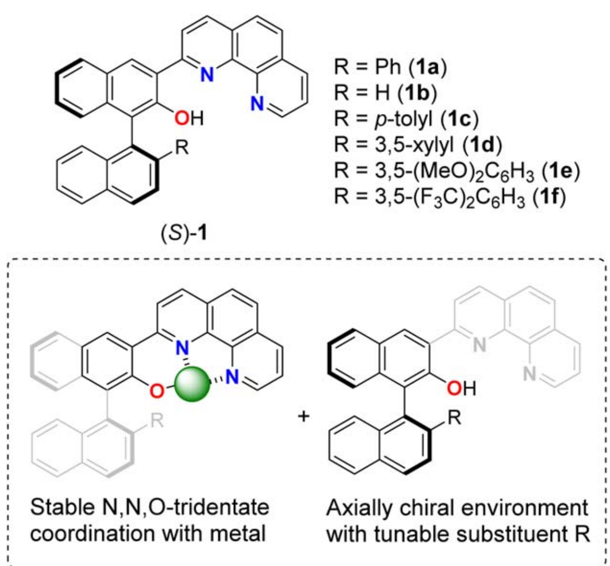


Fig. 4. Axially chiral *N*,*N*,*O*-tridentate phen ligand (*S*)-**1**.

3. Application in Zn Catalysis

In the evaluation of new chiral (*S*)-**1**, we kept in mind that a very limited range of metals have been applied to enantioselective reactions catalyzed by chiral phen ligands as mentioned above,[5–9] in sharp contrast to the non-asymmetric reactions where a variety of metal catalysts are used.[15] We expected ligand (*S*)-**1** would form the corresponding complexes with a wide range of metals and show unexplored utility in asymmetric catalysis. Among the various metal candidates, we initially investigated the catalytic activity in Zn catalysis.[16] Zinc complexes of phen are also reported in inorganic chemistry; however, the organic reactions catalyzed by such complexes have been limited.[17]

3.1. Enantioselective 1,2-Addition of Organozinc Reagents to Aldehydes

We started our investigation into the catalytic activity of (*S*)-**1** with the model reaction of 1,2-addition of organozinc reagents

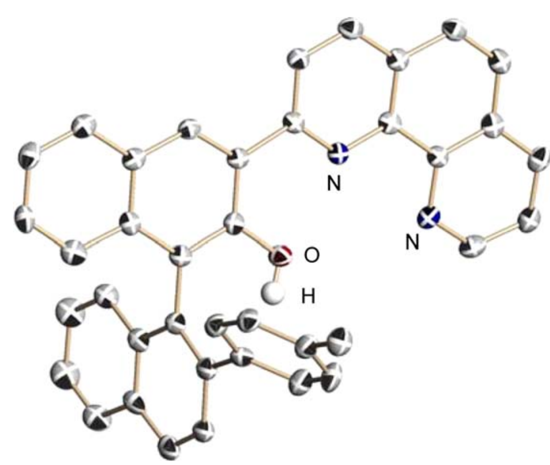
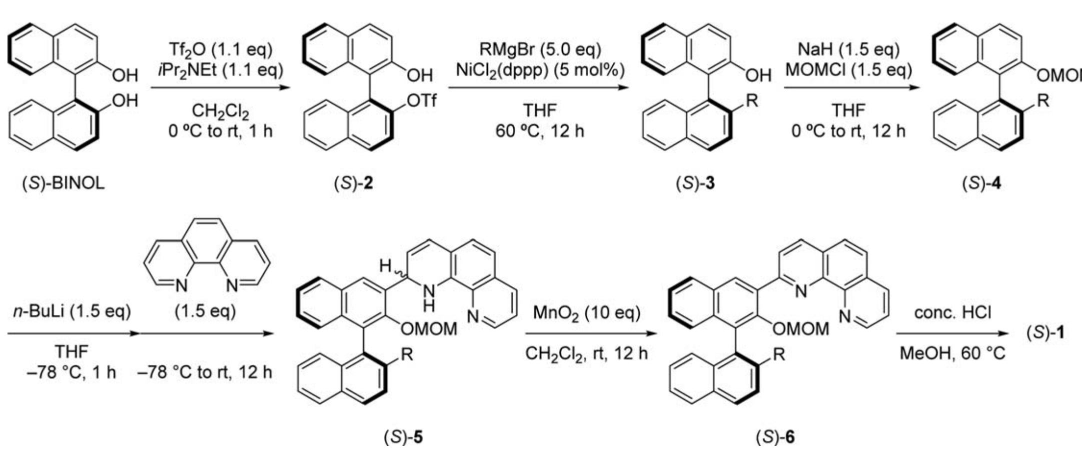
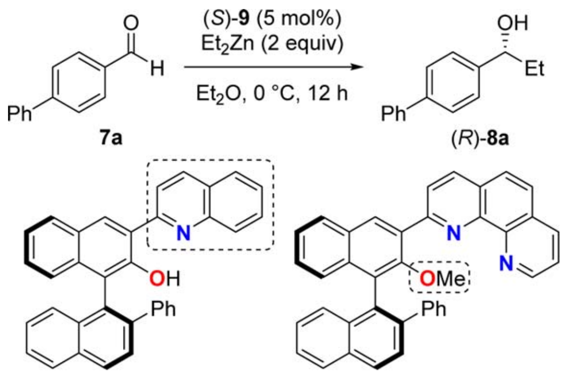


Fig. 5. Molecular structure of (*S*)-**1c**. Hydrogen atoms except that of the OH group are omitted for clarity.



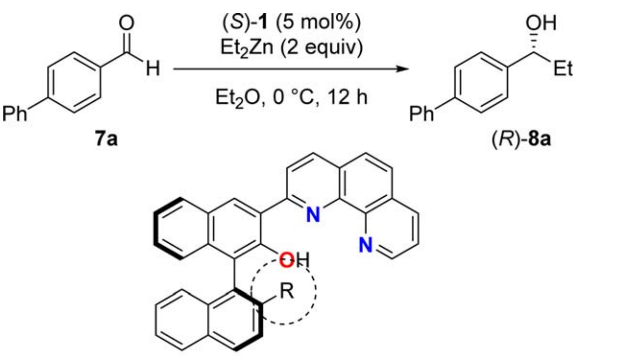
Scheme 2. Synthesis of chiral phen (*S*)-**1**.



9a: 54% yield, –15% ee **9b**: 34% yield, 5% ee

Scheme 3. Control experiments with analogous ligand (*S*)-**9**.

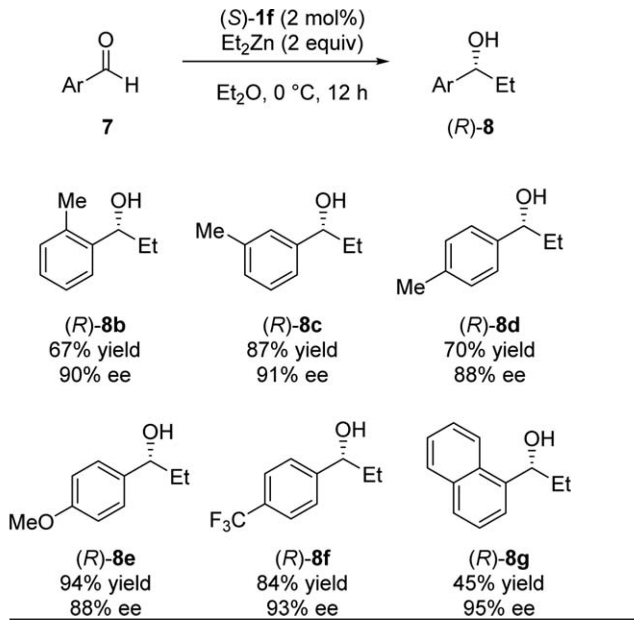
Table 1. Screening of ligands (*S*)-1 in enantioselective addition of diethylzinc to **7a**.



(*S*)-**1**

R = Ph (**1a**): 85% yield, 84% ee
R = H (**1b**): 37% yield, 3% ee
R = *p*-tolyl (**1c**): 99% yield, 85% ee
R = 3,5-xylyl (**1d**): 98% yield, 87% ee
R = 3,5-$(MeO)_2C_6H_3$ (**1e**): 44% yield, 86% ee
R = 3,5-$(F_3C)_2C_6H_3$ (**1f**): **99% yield, 90% ee**

Table 2. Scope of enantioselective addition of diethylzinc to **7**.



to aldehydes, which is an important C–C bond formation reaction providing optically active secondary alcohols. Although a variety of chiral ligands for the enantioselective addition of organozinc reagents have been discovered so far, the employment of phen derivatives has been rarely investigated, even achiral ones.[18] On the other hand, *N*,*N*,*O*-tridentate chiral ligands prepared from natural chiral sources were applied to asymmetric organozinc addition.[19]

Scheme 3 summarizes the results of screening (*S*)-**1** in the model reaction of diethylzinc and *p*-phenylbenzaldehyde **7a** in Et_2O at 0°C (Table 1). The study of substituent effects in (*S*)-**1** revealed that the presence of an aromatic ring as substituent R of the naphthyl ring is very important to obtain the product (*R*)-**8a** in good yield with high enantioselectivity. For example, the use of ligand (*S*)-**1a** with a phenyl group resulted in the formation of (*R*)-**8a** in 85% yield and 84% *ee*. On the other hand, the reaction in the presence of ligand (*S*)-**1b** without any aromatic group led to the dramatic decrease of both yield and *ee* (37% yield, 3% *ee*). Therefore, further study on the substituent effect of the phenyl group of (*S*)-**1a** determined that (*S*)-**1f** with the 3,5-bis(trifluoromethyl)phenyl group was the most effective ligand.

Selected substrate scope is shown in Table 2. Regardless of the position of the substituent on the aromatic ring, the reactions of *o*-, *m*-, and *p*-tolualdehydes uniformly furnished the corresponding alcohols (*R*)-**8b–d** with high enantioselectivity. Concerning the electronic effects in the substrate, both *p*-anisaldehyde and *p*-(trifluoromethyl)benzaldehyde were tolerated. The reaction of 1-naphthaldehyde gave the product (*R*)-**8g** with the highest *ee* of 95%.

In control experiments, we observed that the analogous bidentate ligands (*S*)-**9a** and **9b** gave poorer results and the monoanionic *N*,*N*,*O*-tridentate property of (*S*)-**1** was essential to obtain high enantioselectivity (Scheme 3).

A plausible mechanism for the asymmetric induction is illustrated in Figure 6. Since ligand (*S*)-**1** displays C_1 symmetry, the orientation of carbonyl coordination should be distinguished as in **I-1** and in **I-2**. We assume that the more favored intermediate should be **I-1** due to the steric repulsion between the aromatic ring of (*S*)-**1** and the aldehyde in the case of **I-2**. The nucleophilic addition occurs from the *Re* face of the carbonyl carbon to give (*R*)-**8a** as a major enantiomer. This plausible mechanism clearly matches the fact that (*S*)-**1b** without any aromatic rings did not function (Table 1).

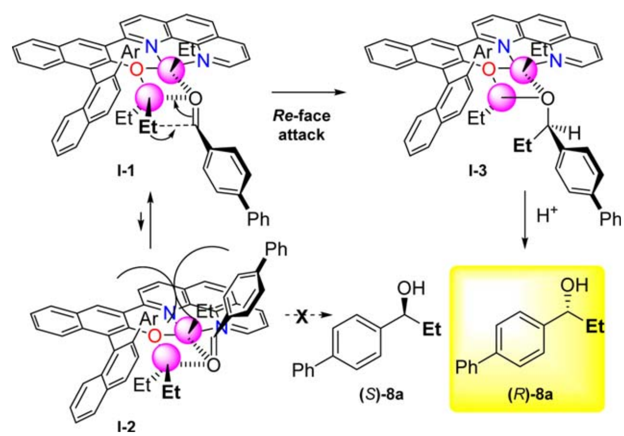


Fig. 6. Plausible mechanism of addition of diethylzinc.

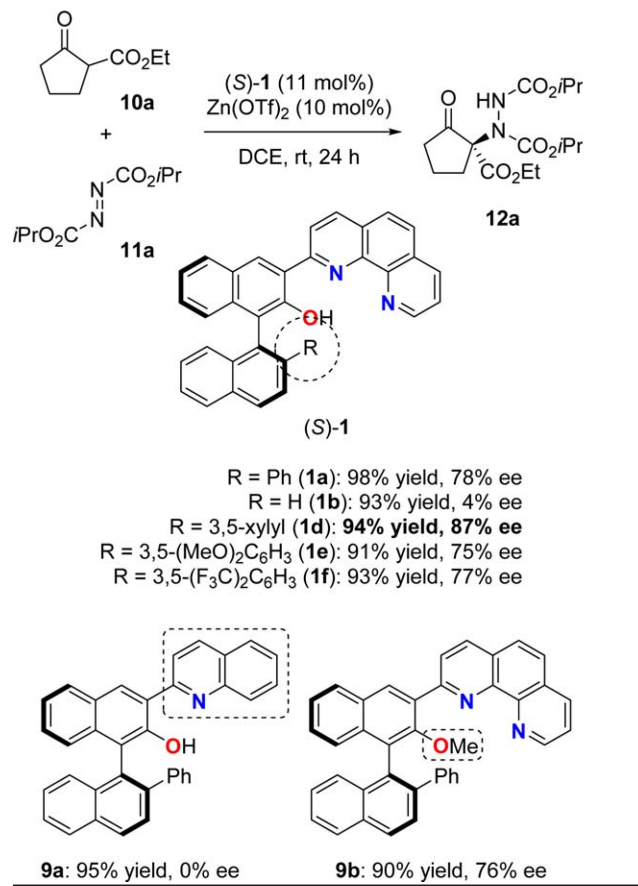
3.2. Enantioselective Amination of *β*-Keto Carbonyl Compounds with Azodicarboxylate

Encouraged by the enantioselective addition of organozinc reagents, we investigated the utility of the Zn(II) complex of (*S*)-**1** as a chiral Lewis acid catalyst. The screening of several model reactions revealed that Zn(II)-catalyzed electrophilic amination of *β*-ketocarbonyl compounds with azodicarboxylates[20,21] gave a promising result. We selected ethyl 2-oxocyclopentanecarboxylate **10a** as a model substrate and conducted the amination reaction of **10a** with diisopropyl azodicarboxylate **11a** in the presence of complexes of (*S*)-**1** and $Zn(OTf)_2$ in DCE for 24 h at room temperature. The study on the effect of ligand structure is shown in Table 3. Also in this case, the substituent R on the naphthyl ring of (*S*)-**1** had a significant impact on the *ee* value. Chiral phen (*S*)-**1d** with a 3,5-xylyl group was the best ligand, giving the desired product **12a** with 87% *ee*.

Also in this reaction, we performed control experiments using control ligands (*S*)-**9a** and **9b** (Table 3). The reaction with (*S*)-**9a** led to a complete loss of enantioselectivity, probably due to the weaker coordination ability of (*S*)-**9a** than (*S*)-**1**. The reaction with (*S*)-**9b** gave the corresponding product **12a** with a similar enantiomeric excess as (*S*)-**1a** (76% *ee* vs 78% *ee*), whereas (*S*)-**9b** was an ineffective chiral ligand in the enantioselective addition of organozinc reagents. The present result suggests that the presence of a free proton within (*S*)-**1** does not have significant influence on the asymmetric induction and (*S*)-**1** might work as a neutral ligand toward $Zn(OTf)_2$, even though it is difficult to determine the coordination as *N*,*N*-bidentate or *N*,*N*,*O*-tridentate.

The selected scope of amination using the optimized azodicarboxylate **11b** is shown in Table 4. The reactions of several five-membered cyclic *β*-ketoesters **10b–e** provided **12b–e** with *ee* values of 90–93%. The ring size of **10** affected the enantioselectivity. For example, six-membered and seven-membered products **12f** and **12g** were obtained with lower selectivity.

Table 3. Screening of (*S*)-1 in the Zn(II)-catalyzed amination of **10a**.



4. Application in Cu Catalysis

To expand the utility of (*S*)-**1**, we wished to make clear the scope of its use in other transition-metal catalysis. In this vein, we tried to understand the structure of metal complexes with (*S*)-**1** by spectrographic analysis. Only one example of a metal complex with *N*,*N*,*O*-tridentate ligand based on the combination of phen backbone and phenol derivatives was found in the literature. Stack et al. synthesized and characterized achiral phen-phenolate Cu(II) complexes **13** as a chemical model of galactose oxidase (Scheme 4).[22] Even though such Cu(II) complexes should have promising properties for alcohol oxidation, further studies on the catalytic activity toward organic synthesis have been limited.

Owing to the abundance of optically active α-hydroxy carbonyl compounds in nature, the asymmetric introduction of a hydroxyl group at the α-carbon of carbonyl compounds has gained considerable attention in asymmetric synthesis.[23] Treatment of a metal enolate with *N*-sulfonyloxaziridine (Davis's oxaziridine) offers a versatile route to nucleophilic α-hydroxylation of carbonyl compounds.[24] Several research

Table 4. Scope of Zn(II)-catalyzed enantioselective amination of **10**.

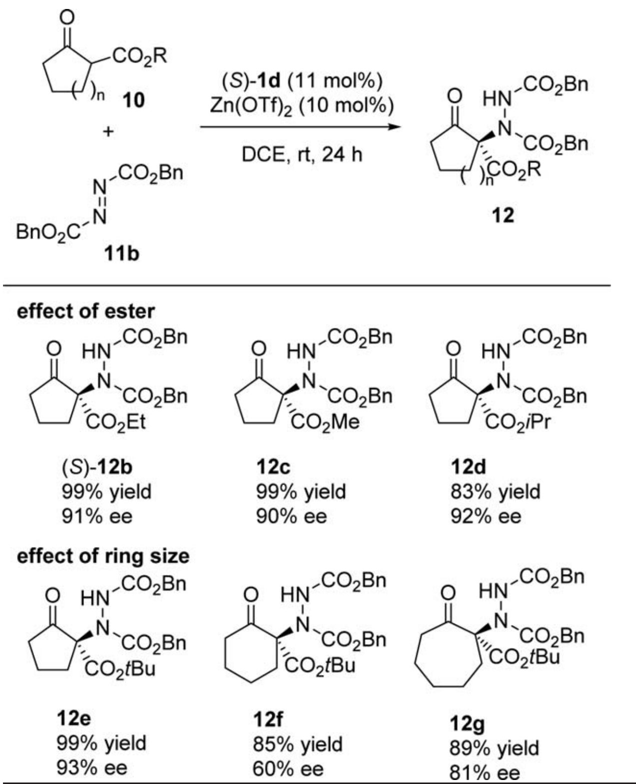
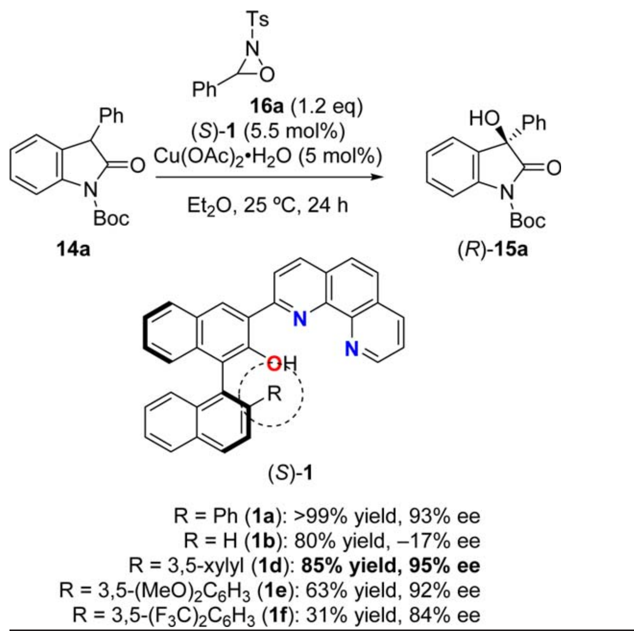
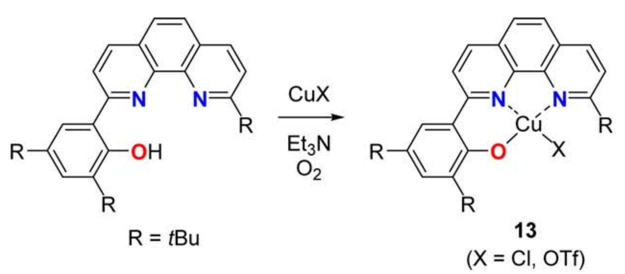


Table 5. Screening of (*S*)-1 in the Cu(II)-catalyzed Davis oxidation of **14a**.





Scheme 4. Stack's synthesis of Cu(II) complex **13**.

groups have designed their own chiral ligands for metal-catalyzed enantioselective α-hydroxylation of carbonyl compounds using Davis's oxaziridine (Davis oxidation).[25] In this context, we applied chiral phen ligands (*S*)-**1** to Cu(II)-catalyzed enantioselective Davis oxidation.[26]

We selected the Davis oxidation of *N*-Boc-3-phenyloxindole (**14a**) with Davis's oxaziridine **16a** in the presence of the Cu(II) complex of (*S*)-**1** generated in situ (Table 5). The optimization of reaction conditions revealed that the reaction with the complex of (*S*)-**1d** prepared from copper(II) acetate monohydrate in Et_2O gave the hydroxylated product (*R*)-**15a** in 85% yield with 95% *ee*.

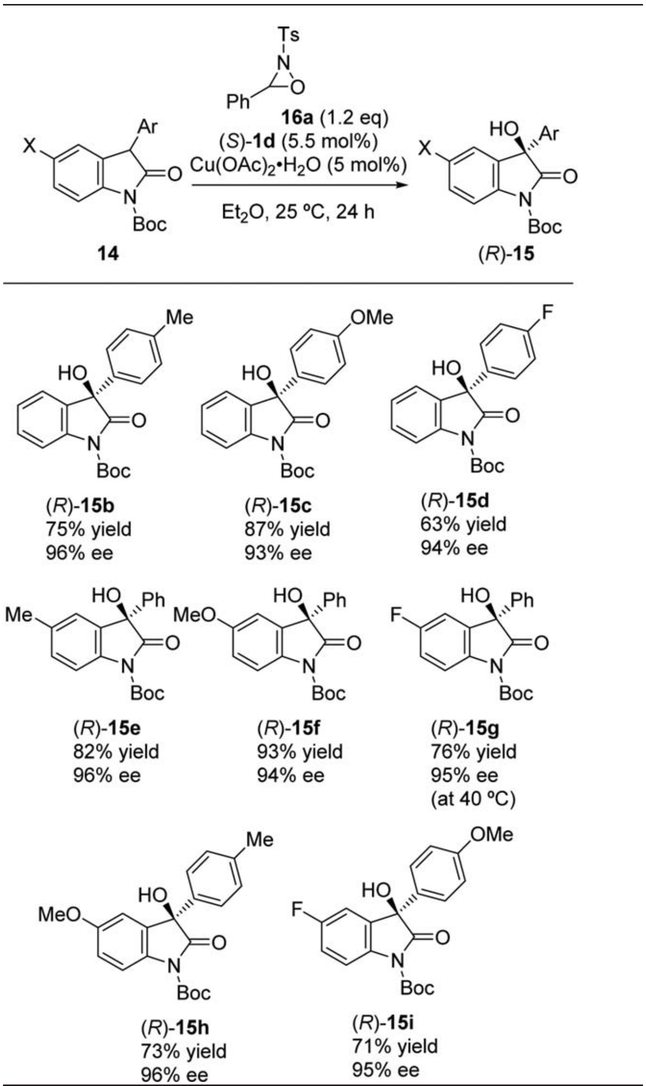
The selected scope of this Cu(II)-catalyzed Davis oxidation of **14** is summarized in Table 6. The oxindoles **14b–d** with a variety of *para*-substituted aromatic rings were all tolerated to give the corresponding products (*R*)-**15b–d** regardless of the electronic factor. Concerning the substituents on the oxindole backbones, the reactions of **14e–i** gave the desired products (*R*)-**15e–i** with uniformly excellent *ee* values.

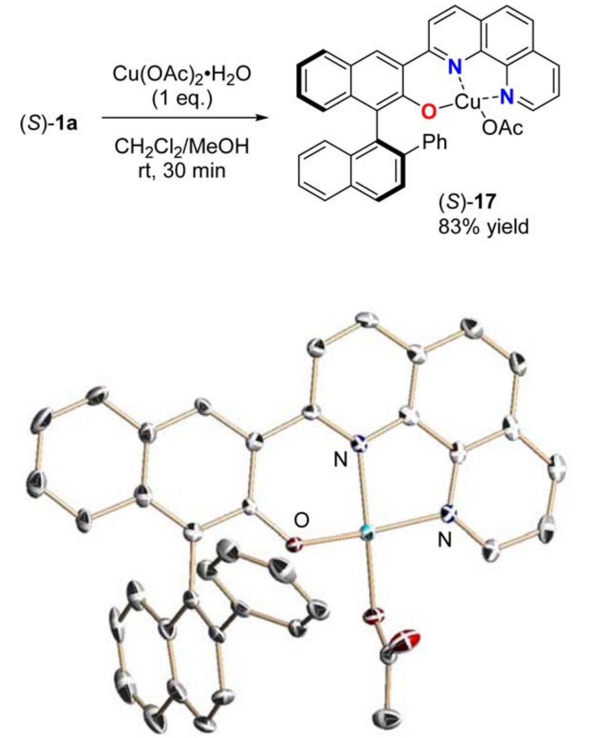
We isolated the Cu(II) complex (*S*)-**17**, which was prepared by the reaction of copper(II) acetate monohydrate and (*S*)-**1a** in a mixed solvent of CH_2Cl_2 and MeOH (Scheme 5). We conducted X-ray crystallographic analysis of (*S*)-**17** to confirm the *N*,*N*,*O*-tridentate structure.

A reaction with this isolated Cu(II) complex (*S*)-**17** was carried out (Scheme 6). Although a slightly poorer result was obtained (50% yield, 89% *ee*), we found that the addition of a catalytic amount of AcOH improved the yield and *ee* to the level of the reaction with the Cu(II) complex prepared in situ (Scheme 6). This result shows that acetic acid, generated from (*S*)-**1a** and copper acetate monohydrate, activated Davis's oxaziridine **16a** to enhance the electrophilicity.

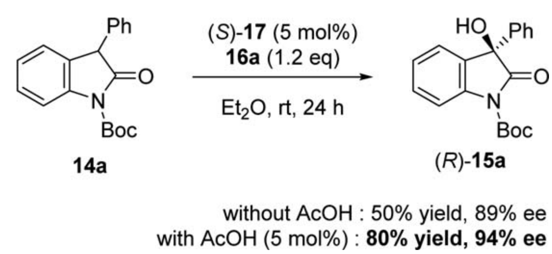
The proposed explanation of the asymmetric induction is illustrated in Figure 7. We assume that the role of the Boc group on the nitrogen atom of **14** is not only as a protecting group but also as a bidentate coordinative group with the Cu(II) center. (*S*)-**1** shows C_1 symmetry, and the Cu(II) enolate intermediates are distinguished as ***I-1*** and ***I-2***. We hypothesized that the more favored intermediate is ***I-1*** to avoid steric repulsion between the bulky Boc group and the aromatic ring of (*S*)-**1** in ***I-2***. The naphthyl ring of (*S*)-**1** blocks the *Si* face of the Cu(II) enolate and the oxaziridine **16a** would approach from the *Re* face to furnish (*R*)-

Table 6. Scope of Cu(II)-catalyzed Davis oxidation of oxindole derivatives **14**.





Scheme 5. The preparation of Cu(II) complex (*S*)-**17** and the molecular structure.



Scheme 6. Reaction using the isolated Cu(II) complex (*S*)-**17**.

15 as the major product. This consideration explains why (*S*)-**1b** without an aromatic ring is ineffective as a ligand (Table 5).

5. Application in Ni Catalysis

Our mechanistic study of the Cu(II)-catalyzed enantioselective Davis oxidation of **14** suggests that the asymmetric induction of this hydroxylation reaction should be controlled by the effective shielding of prochiral Cu(II) enolate surrounded by chiral ligand (*S*)-**1** (Figure 7). Based on this consideration, we envisioned that the present catalytic system could be further extended to the use of other electrophiles. We investigated the enantioselective Michael addition of **14** to methyl vinyl ketone (MVK) (Scheme 7).[27]

Disappointingly, the reaction of **14a** with the Cu(II) complex of (*S*)-**1a** gave the product **18a** in a low yield and the racemic form. The subsequent screening of metal precursors revealed that only the reaction with the complex of nickel(II) acetate tetrahydrate and (S)-**1a** gave (*R*)-**18a** in quantitative yield and with moderate enantioselectivity.[28] To further increase the *ee* value, we introduced an additional effective chiral environment within (*S*)-**1a**. We anticipated that the introduction of the additional phenyl substituent within the phen backbone of (*S*)-**1a** would enhance steric factors. Gratifyingly, the reaction with this modified ligand (*S*)-**1g** gave the corresponding Michael adduct (*R*)-**18a** in 92% yield and with 87% *ee*. The selected scope of the Michael addition of **14** to MVK is shown in Table 7.

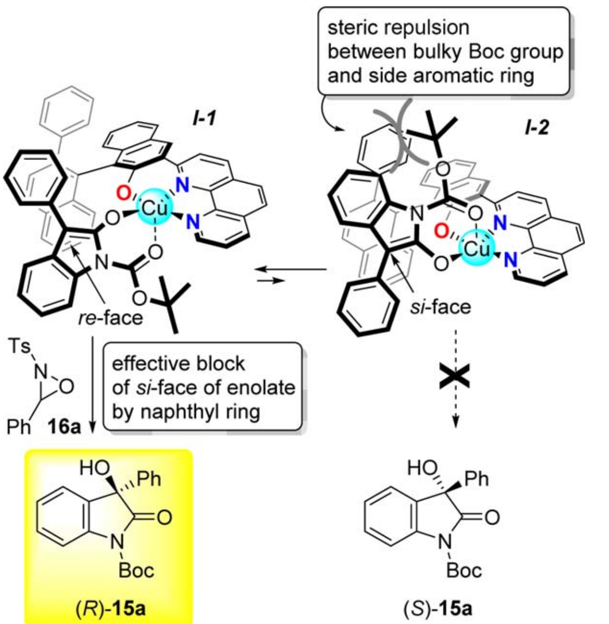
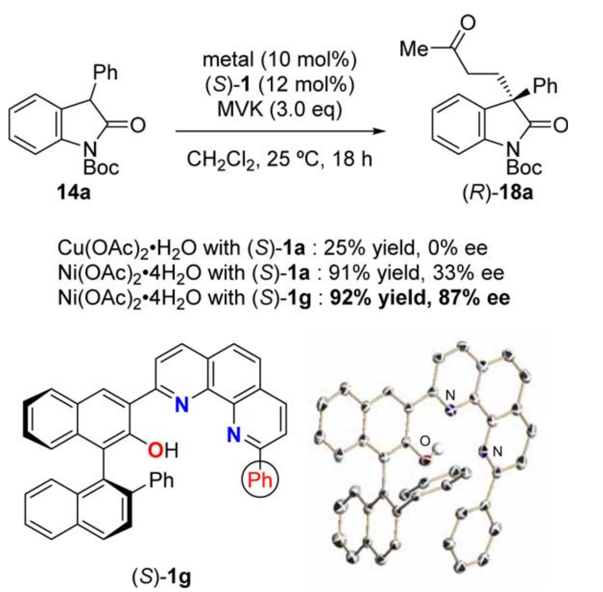


Fig. 7. Plausible explanation of asymmetric induction in Davis oxidation of **14**.

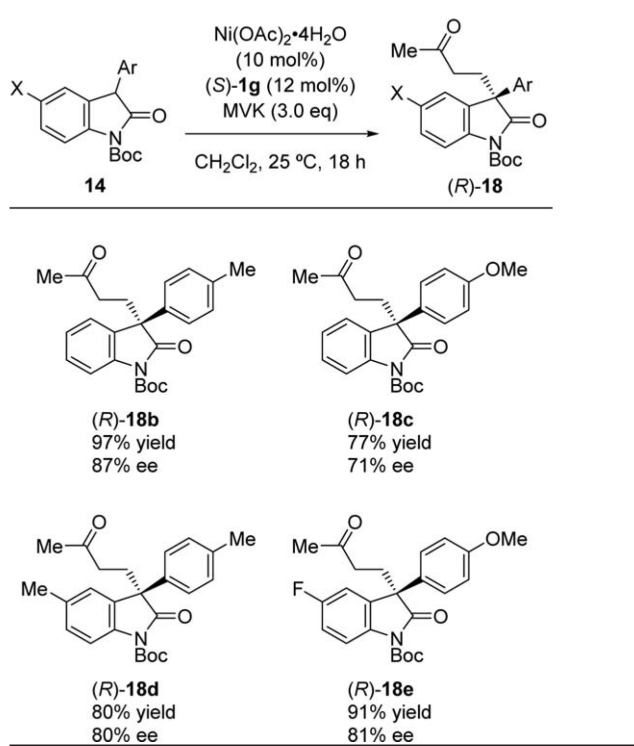


Scheme 7. The Michael addition of **14a** to MVK with the Ni(II) complex of the modified ligand (*S*)-**1g**.

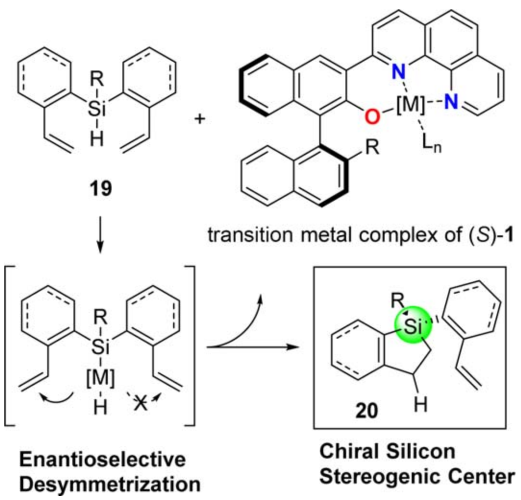
6. Application in Rh Catalysis

We have succeeded in demonstrating the catalytic activity of (*S*)-**1** in chiral Lewis acid catalysis of Zn(II), Cu(II), and Ni(II) for enantioselective C–C and C–heteroatom bond formations. The following interest in the application of (*S*)-**1** focused on the use of unprecedented transformations using transition-metal catalysis involving the redox cycle. In this chapter, we discuss the Rh-catalyzed enantioselective desymmetrization of hydrosilanes towards the construction of chiral silicon stereogenic centers,[29] which have been of great interest considering their unique chemical and physical properties. In this context, the aim of our synthetic plan is the utilization of the enantioselective desymmetrization of prochiral silanes **19** via intramolecular hydrosilylation in the presence of a chiral transition-metal catalyst, thus giving chiral silicon-containing olefins **20** (Scheme 8).[30] A few examples of the catalytic asymmetric preparation of chiral silicon compounds utilizing hydrosilylation protocols were reported by Tamao and Tomooka.[31]

Table 7. Scope of Ni(II)-catalyzed Michael addition of **14**.

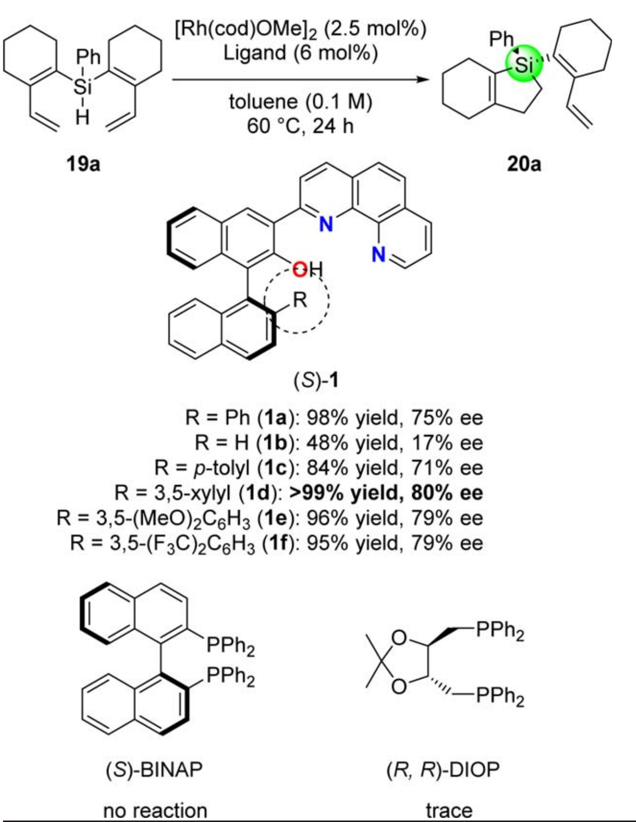


To identify effective catalytic systems, we performed the model reaction of hydrosilane **19a** with various transition-metal complexes of (*S*)-**1** in toluene at 60°C (Table 8). $[Rh(cod)Cl]_2$ was found to be an effective transition-metal complex of (*S*)-**1a** to give the corresponding five-membered cyclic organosilane **20a** in moderate yield and with good *ee* (53% yield, 70% *ee*), whereas the use of $Pt(dba)_2$, $Ni(cod)_2$, or $Pd(dba)_2$ resulted in no reaction. The screening of Rh(I) salts demonstrated that $[Rh(cod)OMe]_2$ was the most effective precursor to provide **20a** in quantitative yield and with 75% *ee*. The reaction with (*S*)-**1d** with the 3,5-xylyl group gave the



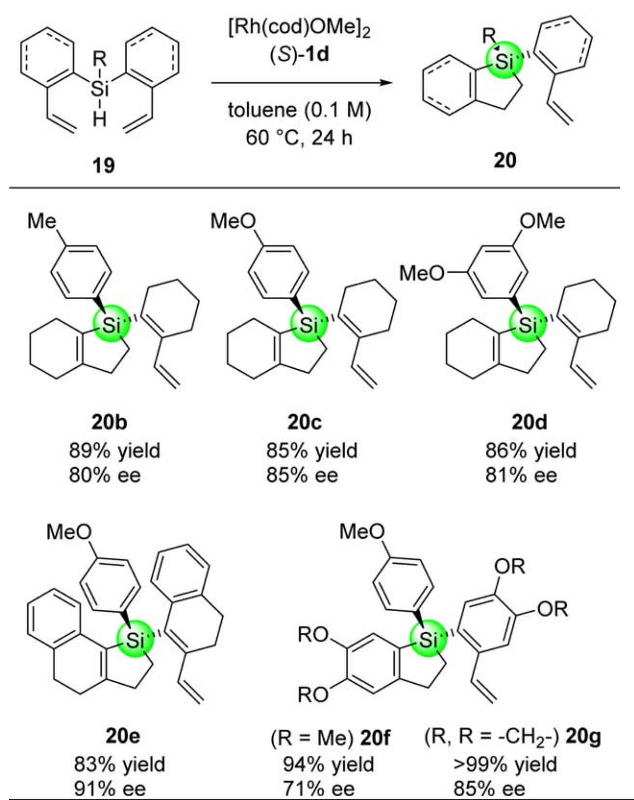
Scheme 8. Concept of enantioselective desymmetrizing hydrosilylation.

Table 8. Screening of (*S*)-1 in the Rh(I)-catalyzed desymmetrizing hydrosilylation of **19a**.



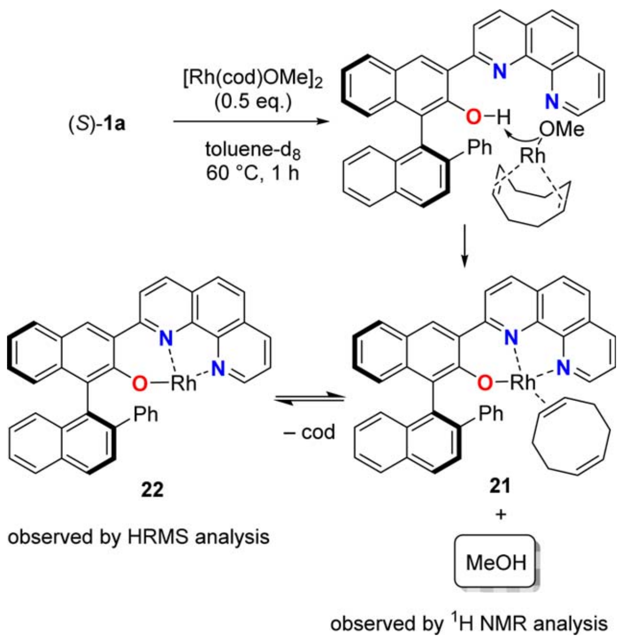
best *ee* (>99% yield, 80% *ee*). It should be noted that the reactions with commercially available common chiral phosphine ligands such as (*S*)-BINAP and (*R*,*R*)-DIOP were not successful.

Table 9. The scope of Rh(I)-catalyzed enantioselective desymmetrization of **19**.



The selected scope of this Rh(I)-catalyzed enantioselective desymmetrization of hydrosilane **19** is summarized in Table 9.

On the basis of the result that a simple *N*,*N*-bidentate phen was an ineffective ligand for this reaction, we assumed that the chiral Rh(I) complex of (*S*)-**1** generated in situ adopts *N*,*N*,*O*-tridentate coordination. We analyzed Rh(I) complex **21**, prepared from the reaction of $[Rh(cod)OMe]_2$ and (*S*)-**1a** in toluene-d_8 at 60°C for 1 h (Scheme 9). We assigned the *N*,*N*,*O*-tridentate coordinated structure by ^1H NMR analysis. Strong evidence for this *N*,*N*,*O*-tridentate coordination was given by the observation of the generation of MeOH, which suggests that the proton of the hydroxy group of (*S*)-**1a** was removed by the basic methoxide anion of $[Rh(cod)OMe]_2$. ^1H NMR analysis also indicated that one olefin of the external COD ligand coordinated with the Rh(I) center and the other one was a free ligand. The monodentate COD should be readily dissociated from the Rh(I) center and hydrosilane **19** can approach this vacant site. We also observed the presence of **22**, which had undergone release of the COD ligand from **21**, by HRMS analysis.



Scheme 9. The possible structures of *N*,*N*,*O*-tridentate Rh(I) complexes **21** and **22**.

7. Summary and Outlook

1,10-Phenanthroline is definitely a powerful chelating reagent and it has been applied to a broad range of academic research areas. In contrast, the utilization of this highly coordinative ligand as a structural motif for chiral ligands has remained a frontier in the field of asymmetric synthesis. We introduced the axially chiral *N*,*N*,*O*-tridentate phenanthroline ligand (*S*)-**1**. The design of (*S*)-**1** is based on the concept of renovation of optically active phen derivatives as chiral ligands efficient for versatile asymmetric metal catalysis. Our endeavor to develop enantioselective transformations has been devoted to several enantioselective reactions such as (i) addition of organozinc reagents to aldehydes,[11] (ii) Zn(II)-catalyzed electrophilic amination of β-ketoesters with azodicarboxylate,[21] (iii) Cu(II)-catalyzed α-hydroxylation of oxindole derivatives,[26] (iv) Ni(II)-catalyzed Michael addition of oxindole derivatives to MVK,[27] and (v) Rh(I)-catalyzed intramolecular hydrosilylation for the construction of chiral silicon stereogenic centers.[30] Further applications using other metal catalysts are ongoing to broaden the utility of (*S*)-**1**. Furthermore, we are attempting to realize the unique elaboration of the structural features of (*S*)-**1**, for example, (i) cooperative catalysis with *N*,*N*-bidentate metal complexes and intramolecular hydrogen bonding provided by phenolic hydroxyl groups and (ii) the preparation of hetero-bimetallic complexes of (*S*)-**1** having transition-metal ions coordinated with the *N*,*N*-bidentate phen backbone and oxophilic metals trapped by phenolic hydroxyl groups. We believe that the chiral phen ligands (*S*)-**1** could display a broad utility for more varied metal catalysts and transformations in future investigations.

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